Exp 4: Develop a simple calculator using XMLRPC.

Aim:

To develop a simple calculator application using XML-RPC (Remote procedure call) in Python (implementing a client-server architecture)

Procedure:-

1) Create server script 'calc_server.py' using simple XMLRPC server.

2) Define arithmetic functions: add, subtract, multiply, divide, power, modulo.

3) Register all functions using server.register_introspection_functions().

4) Start the server with server.serve_forever() on port 8000.

5) Connect to the server URL 'http://localhost:8000'.

6) Call remote functions through the proxy object.

OUTPUT:

```
def add (a,b) : return a+b
def subtract (a,b) : return a-b
def Multiply (a,b) : return a*b
def divide (a,b):
    if b== 0:
    return a/b.

Print (f'10 +5 = {Proxy.add (10,5)}')
Print (f'10-5 = {proxy.Subtract(10,5)}')
Print (f'10*5 = {proxy.Multiply(10,5)}')
Print (f'10/5 = {proxy.Power (2,8)}')
Print (f'10%3 = {proxy.Modulo(10,3)}').
```

7) Display the results returned from the server.

8) Test all arithmetic operations.

9) Handle division by zero and invalid input exceptions

Result:

The XML-RPC Calculator application was successfully developed. The server exposed arithmetic operations and the client called them remotely. All mathematical operations were verified correctly.